NLPrompt: Noise-Label Prompt Learning for Vision-Language Models

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Abstract

Prompt learning is an efficient way to adapt large vision-language models such as CLIP, but the standard cross-entropy objective remains brittle when few-shot supervision is corrupted by label noise. We present NLPrompt, a simple robust prompt-learning framework that combines a noise-tolerant prompt loss with optimal-transport-based sample purification. The first component, PromptMAE, replaces cross-entropy with mean absolute error, implemented as generalized cross-entropy with $q = 1$, which suppresses gradients from examples whose assigned labels are inconsistent with the current model. The second component, PromptOT, uses class text features induced by the learned prompts as prototypes, constructs an entropic optimal transport assignment between image and text features, and uses the resulting coupling to separate trusted and noisy training samples. NLPromptthen optimizes cross-entropy on the trusted subset and MAE on the remaining subset. Across seven synthetic noisy-label benchmarks and the real-world Food101N benchmark, NLPromptconsistently improves robustness, especially at high noise rates. For example, on Flowers102 with 75.0% symmetric noise it reaches 76.80% accuracy versus 70.37% for GCE, 66.93% for JoAPR, and 37.17% for CoOp, while on Food101N it achieves 76.46%, outperforming JoAPR by 3.89 points. Additional ablations show that both the MAE branch and OT-based purification are necessary, and that the proposed training recipe transfers to VPT, MaPLe, and PromptSRC.

1. Introduction

Large vision-language models (VLMs), typified by CLIP, have made prompt-based adaptation an attractive alternative to full fine-tuning. Methods such as CoOp, CoCoOp [30], MaPLe [14], VPT [13], and PromptSRC [15] adapt pretrained representations by learning a small set of task-specific tokens while keeping most backbone parameters frozen. This parameter efficiency is especially valuable in few-shot settings, where prompt learning can leverage the pretrained alignment between image and text spaces.

The remaining bottleneck is supervision quality. Real downstream datasets often contain label corruption from automatic collection, weak annotation, or class ambiguity. Under noisy labels, standard prompt learning inherits the failure mode of conventional cross-entropy training: once the model starts to fit corrupted labels, the prompt tokens are pulled away from the stable image-text geometry learned during pretraining. This problem is particularly severe in the few-shot regime because each corrupted example occupies a large fraction of the available supervision. Existing analyses show that prompt tuning is more noise-tolerant than end-to-end fine-tuning because the frozen class tokens and pretrained alignment regularize optimization [28], but the results in Table 1 show that the default cross-entropy objective still degrades sharply as noise increases.

This paper studies a simple question: can prompt learning become substantially more noise-robust without sacrificing its efficiency or clean-data behavior? Our answer is yes. We observe that two ingredients are sufficient. First, the loss on suspicious samples should attenuate gradients from likely label mismatches. Second, the cleanest portion of the data should still benefit from the discrimination of cross-entropy. These observations lead to NLPrompt, which combines PromptMAE and PromptOT. PromptMAE uses mean absolute error in probability space, implemented as generalized cross-entropy with $q = 1$, to suppress outlier gradients. PromptOT uses prompt-induced text features as class prototypes and computes a global entropic optimal transport plan between training images and classes, producing pseudo-labels and a budget-controlled sample split. The trusted subset is optimized with cross-entropy, while the remainder is optimized with MAE.

The resulting method is simple, fully compatible with frozen-backbone prompt learning, and effective across both synthetic and real noisy-label settings. On the severe 75.0% symmetric-noise benchmark, NLPromptimproves over the strongest listed baseline by 6.43 points on Flowers102, 8.40 on DTD, 12.40 on EuroSAT, 3.90 on OxfordPets, 7.43 on StanfordCars, 2.86 on UCF101, and 2.77 on Caltech101. On asymmetric noise it remains strongest in nearly all medium- and high-noise regimes, while Food101N confirms the gains under naturally corrupted labels.

Our contributions are threefold. First, we show that

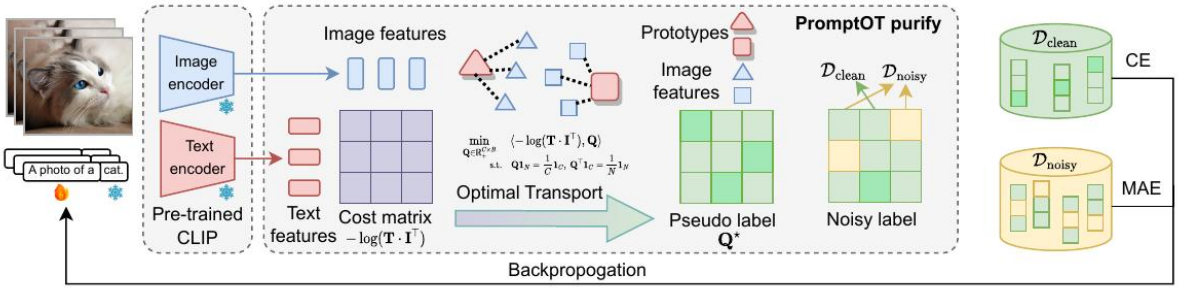

Figure 1. **Overview of NLPrompt.** The prompt learner produces class text prototypes from CLIP's frozen text encoder. For each epoch, image features and prompt-induced text features are matched by entropic optimal transport to obtain pseudo-labels and a trusted-sample mask. Samples whose noisy label agrees with the OT pseudo-label are treated as clean and optimized with cross-entropy; the remaining samples are optimized with MAE. This design preserves the discrimination of cross-entropy on reliable samples while suppressing the influence of likely mislabeled ones.

MAE is a strong loss for robust prompt tuning and explain its behavior through exact gradient attenuation. Second, we introduce an OT-based prompt purification module that uses text prototypes rather than sample-sample heuristics to identify trusted examples. Third, we demonstrate a harmonized clean/noisy training recipe that delivers state-of-the-art robustness across the provided noisy-label benchmarks and transfers to multiple prompt-learning backbones.

2. Related Work

Prompt learning for VLM adaptation. CLIP established image-text contrastive pretraining as a strong transferable foundation. CoOp showed that prompt contexts can be learned directly for downstream recognition, and CoCoOp [30], MaPLe [14], VPT [13], PromptSRC [15], PLOT [2], and ProGrad [31] expanded prompt learning with conditional prompts, multimodal prompts, visual prompts, self-regularization, OT-based prompt matching, and gradient regularization. These methods mainly target clean supervision or generalization shift. NLPromptinstead targets label corruption during prompt tuning and can be layered on top of existing prompt learners.

Learning with noisy labels. Robust learning under label noise has been studied through robust losses [8, 26], correction with estimated transition matrices [23], trusted-data calibration [11], sample selection [3, 22], and semi-supervised partitioning [18]. These methods are usually designed for standard image classifiers and do not directly exploit the text branch of a VLM. NLPromptadapts the noisy-label problem to prompt tuning by using prompt-induced text features as a structured label space and by pairing a robust loss with a prompt-specific purification step.

Optimal transport for robust recognition. Entropic optimal transport offers a global assignment mechanism that avoids making decisions independently for each sample [5]. OT has been used for prompt learning [2], noisy-label denoising [1, 7, 29], and pseudo-label refinement [19, 25]. Our PromptOT is closest in spirit to OT-based label cleaning, but it uses CLIP text prototypes induced by the current prompts, allowing the purification process to stay aligned with the prompt learner's evolving decision space.

Noisy labels in prompt tuning. Wu et al. [28] showed that prompt tuning is intrinsically more robust to noisy labels than full model tuning, yet there remains substantial degradation at high noise rates. JoAPR [9] improves prompt learning by cleaning corrupted supervision through agreement-based regularization. NLPromptdiffers in two ways: it explicitly changes the optimization rule on suspicious samples through MAE, and it uses an OT coupling with prompt-derived text prototypes to obtain a global clean/noisy split.

3. Method

3.1. Problem Setup

We consider few-shot prompt learning for a frozen VLM. Given an image x_i with noisy label $\tilde{y}_i \in \{1, \dots, C\}$, the prompt learner outputs one prompt per class, and the frozen CLIP encoders produce normalized image features $\boldsymbol{v}_i$ and text features $\boldsymbol{t}_j$. The classification logits are

$$s_{ij} = \tau\, \boldsymbol{v}_i^\top \boldsymbol{t}_j, \tag{1}$$

where τ is the learned CLIP logit scale. Only the prompt parameters are updated; the image encoder, text encoder, and class-name tokens remain frozen.

3.2. PromptMAE

The standard CoOp-style objective applies cross-entropy to the noisy label $\tilde{y}_i$:

$$\mathcal{L}_{\mathrm{CE}}(x_i, \tilde{y}_i) = -\log p_{i,\tilde{y}_i}, \tag{2}$$

where $p_i = \mathrm{softmax}(s_i)$. Under label corruption, CE can still assign large gradients to examples whose provided labels strongly disagree with the model.

We therefore use mean absolute error in the class-probability space for noisy samples. In the released implementation this is instantiated as generalized cross-entropy with $q = 1$, namely

$$\mathcal{L}_{\mathrm{MAE}}(x_i, \tilde{y}_i) = 1 - p_{i,\tilde{y}_i}, \tag{3}$$

which is equivalent to multiclass MAE up to a constant factor for one-hot targets. The key property is that the loss decreases the influence of samples whose assigned labels already look implausible under the current prompt.

Proposition 1. *Let $\tilde{y}$ be the provided label for an example with softmax probabilities p_k. Then*

$$\left|\frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial s_{\tilde{y}}}\right| = 1 - p_{\tilde{y}}, \qquad \left|\frac{\partial \mathcal{L}_{\mathrm{MAE}}}{\partial s_{\tilde{y}}}\right| = p_{\tilde{y}}(1 - p_{\tilde{y}}), \tag{4}$$

and for $k \neq \tilde{y}$,

$$\left|\frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial s_k}\right| = p_k, \qquad \left|\frac{\partial \mathcal{L}_{\mathrm{MAE}}}{\partial s_k}\right| = p_{\tilde{y}} p_k. \tag{5}$$

Proof. For CE, $\partial \mathcal{L}_{\mathrm{CE}}/\partial s_k = p_k - \mathbb{1}[k = \tilde{y}]$. For $\mathcal{L}_{\mathrm{MAE}} = 1 - p_{\tilde{y}}$, differentiating the softmax gives $\partial \mathcal{L}_{\mathrm{MAE}}/\partial s_k = -\partial p_{\tilde{y}}/\partial s_k$, which equals $-p_{\tilde{y}}(1 - p_{\tilde{y}})$ for $k = \tilde{y}$ and $p_{\tilde{y}} p_k$ otherwise. □

This proposition is exact and explains why MAE is attractive under noise: when the model assigns very low probability to the provided label, MAE scales the entire gradient by the small factor $p_{\tilde{y}}$, whereas CE still applies nearly unit magnitude on the labeled class. In feature-learning terms, MAE suppresses low-consistency samples and therefore improves the effective signal-to-noise ratio seen by the prompt parameters [12, 27].

3.3. PromptOT

MAE alone is robust but may underuse reliable supervision. We therefore estimate which samples are trustworthy before each epoch. Given current prompt-induced text prototypes $\{\boldsymbol{t}_j\}_{j=1}^{C}$ and image features $\{\boldsymbol{v}_i\}_{i=1}^{N}$, we build an affinity matrix

$$A_{ij} = \tau\, \boldsymbol{v}_i^{\top} \boldsymbol{t}_j. \tag{6}$$

PromptOT computes an entropically regularized OT plan

$$\Pi^{\star} = \arg\max_{\Pi \in \mathcal{U}(\boldsymbol{a}, \boldsymbol{b})} \langle \Pi, A \rangle + \varepsilon H(\Pi), \tag{7}$$

where $\mathcal{U}(\boldsymbol{a}, \boldsymbol{b})$ is the transport polytope with row marginal $\boldsymbol{a}$ and class prior $\boldsymbol{b}$, and $H(\Pi)$ is the entropy term solved by Sinkhorn iterations [5]. In the implementation, $\boldsymbol{a}$ is uniform over training samples and $\boldsymbol{b}$ distributes a configurable budget uniformly across classes, which yields a global confidence-aware selection mask.

From $\Pi^{\star}$, each sample receives a pseudo-label

$$\hat{y}_i = \arg\max_j \Pi^{\star}_{ij}, \tag{8}$$

and a confidence score given by the row-wise maximum of the transport plan. The top-confidence samples selected by the OT budget form a trusted candidate pool. We mark a sample as *clean* if it is selected by OT and its pseudo-label agrees with the provided label, $\hat{y}_i = \tilde{y}_i$. All other samples are treated as noisy. Compared with per-sample thresholding, the OT formulation enforces global consistency across the batch of images and the full class set, echoing the benefits observed in OT-based noisy-label filtering [1, 7].

3.4. Harmonized Clean/Noisy Training

The final training rule combines the two modules. At the start of each epoch, PromptOT partitions the training set into a clean subset $\mathcal{C}$ and a noisy subset $\mathcal{N}$. The prompt learner is then updated with cross-entropy on $\mathcal{C}$ and MAE on $\mathcal{N}$:

$$\min_{\theta_{\mathrm{prompt}}} \sum_{i \in \mathcal{C}} \mathcal{L}_{\mathrm{CE}}(x_i, \tilde{y}_i) + \sum_{i \in \mathcal{N}} \mathcal{L}_{\mathrm{MAE}}(x_i, \tilde{y}_i). \tag{9}$$

The released trainer implements these as two sequential passes over the clean and noisy loaders within each epoch. This asymmetry is deliberate. CE preserves class discrimination when the supervision is likely correct, while MAE prevents suspicious samples from dominating optimization. Figure 1 summarizes the full pipeline.

Two implementation details are worth noting. First, PromptOT uses the current prompt-induced text features rather than fixed hand-crafted templates. This makes purification adaptive to the prompt learner and avoids a mismatch between the denoising module and the classifier being optimized. Second, the released code performs OT globally over the current training set, then physically splits the dataloader into trusted and noisy subsets for the subsequent epoch. In other words, purification is not merely a soft regularizer on the logits; it changes which loss each sample receives.

Algorithm 1 highlights the simplicity of the final method. The only trainable parameters are the prompt tokens, and the only extra computation relative to standard prompt learning is the epoch-level OT assignment used to determine which objective each sample should receive.

Algorithm 1 NLPrompt

Initialize learnable prompt context θ and freeze the CLIP image/text encoders
for epoch $= 1$ to T **do**
  Compute prompt-induced text prototypes $\{\boldsymbol{t}_j\}_{j=1}^{C}$
  Encode all training images to obtain $\{\boldsymbol{v}_i\}_{i=1}^{N}$
  Solve entropic OT between image features and text prototypes
  Obtain pseudo-labels $\hat{y}_i$ and a confidence-based selected set
  Form $\mathcal{C} = \{i : \text{selected and } \hat{y}_i = \tilde{y}_i\}$
  Form $\mathcal{N} = \{1, \ldots, N\} \setminus \mathcal{C}$
  Update θ on $\mathcal{C}$ with cross-entropy
  Update θ on $\mathcal{N}$ with MAE
end for

4. Experiments

4.1. Experimental Setup

We evaluate on seven few-shot benchmarks with synthetic label noise: Flowers102 [20], DTD [4], EuroSAT [10], OxfordPets [21], StanfordCars [16], UCF101 [24], and Caltech101 [6]. We also evaluate on Food101N [17], which provides real noisy web labels and uses the Food101 test split in the released code. Following the provided training scripts, the main setup uses CLIP with an RN50 backbone, 16 shots per class, and prompt learning with frozen image and text encoders. Synthetic corruption is reported under both symmetric and asymmetric noise from 12.5% to 75.0%. We report top-1 accuracy.

We compare with standard prompt tuning (CoOp), GCE-based training, and JoAPR [9]. In the analysis section, we further test whether the proposed clean/noisy harmonization transfers to VPT [13], MaPLe [14], and PromptSRC [15]. All numbers are taken directly from the provided experiment tables.

The code-level configuration is also lightweight. The prompt learner uses 16 context tokens, SGD optimization, cosine learning-rate decay, 200 epochs, and a training batch size of 32 in the default RN50 configuration. The released script sweeps the learning rate over $\{10^{-3}, 6 \times 10^{-4}, 7 \times 10^{-4}\}$ and evaluates multiple random seeds. OT is enabled throughout training, with entropic regularization exposed as a dataset-level hyperparameter. These settings keep the method close to standard CoOp-style training and make the improvement attributable to the loss/splitting design rather than to heavy retraining.

4.2. Main Results on Synthetic Label Noise

Table 1 shows that NLPromptis the strongest method overall and that the gains widen as the corruption rate increases. Under 75.0% symmetric noise, NLPromptimproves over the best baseline by 6.43 points on Flowers102, 8.40 on DTD, 12.40 on EuroSAT, 3.90 on OxfordPets, 7.43 on StanfordCars, 2.86 on UCF101, and 2.77 on Caltech101. The improvements are similarly strong under asymmetric noise, where NLPromptreaches 55.33% on Flowers102, 32.73% on EuroSAT, 39.50% on StanfordCars, and 49.27% on UCF101 at 75.0% noise.

Two patterns are worth highlighting. First, the gap over CoOp becomes dramatic at high noise, confirming that standard CE-based prompt learning still overfits corruption despite the frozen backbone. Second, the margin over GCE and JoAPR is largest in the medium-to-high noise regime, which is exactly where clean/noisy separation matters most. At the same time, the table also shows that NLPromptis not uniformly best in every low-noise case. GCE is slightly stronger on StanfordCars at 12.5% noise and on a few Caltech101 entries, while JoAPR is strongest on OxfordPets with 75.0% symmetric noise and on Caltech101 with 75.0% asymmetric noise. This is consistent with our claim: NLPromptis designed primarily for robust adaptation under severe corruption rather than for squeezing out every last fraction of a point in the nearly clean regime.

The dataset-specific trends are also informative. Flowers102 and EuroSAT exhibit very large gains under both symmetric and asymmetric corruption, suggesting that prompt-induced text prototypes are particularly effective when class semantics are relatively coherent and the corrupted labels remain recoverable from the image-text geometry. StanfordCars and UCF101 are harder, but NLPromptstill maintains a clear advantage once the noise rate exceeds 37.5%. OxfordPets is the main exception: while NLPromptdominates most settings, JoAPR remains stronger at the hardest symmetric-noise point. This suggests that when the trusted set becomes very small and class boundaries stay visually subtle, agreement-based regularization can still be competitive.

4.3. What the Results Say About the Design

The combined table points to a consistent division of labor between the two components. MAE is crucial for avoiding catastrophic collapse at high noise, while OT matters because a fully MAE-trained prompt learner can become too conservative. This is most visible on datasets where GCE remains competitive at low noise but loses rapidly afterward. NLPromptcloses that gap by reserving CE for samples that are globally consistent with the current prompt-induced label geometry.

The asymmetric-noise results are especially important because they are harder to explain away with simple memorization arguments. For example, on UCF101 at 75.0% asymmetric noise, NLPromptreaches 49.27% while JoAPR obtains 47.46% and GCE 36.37%. On Flowers102, the margin is substantially larger: 55.33% versus 39.23% and

Table 1. Performance metrics across various datasets and noise levels. (%)

Dataset	Method	Noise Rate: Sym						Noise Rate: Asym					
		12.5%	25.0%	37.5%	50.0%	62.5%	75.0%	12.5%	25.0%	37.5%	50.0%	62.5%	75.0%
Flowers102	CoOp	88.93	83.50	77.93	70.10	55.60	37.17	86.97	74.70	60.43	42.60	26.53	12.60
	GCE	88.80	88.33	86.73	84.07	78.37	70.37	88.40	86.37	80.33	69.93	61.50	39.23
	JoAPR	85.57	81.23	74.60	70.23	67.90	66.93	85.17	79.63	73.97	73.83	53.37	13.27
	NLPrompt	**93.87**	**92.57**	**92.73**	**89.90**	**84.77**	**76.80**	**93.80**	**93.40**	**91.77**	**81.10**	**73.63**	**55.33**
DTD	CoOp	56.00	49.57	43.30	34.37	27.83	17.27	55.60	47.75	38.07	29.63	20.53	11.70
	GCE	61.00	59.83	56.80	50.73	43.60	33.67	60.70	57.57	52.70	43.97	33.40	18.23
	JoAPR	58.07	57.70	56.33	53.03	48.05	29.90	52.40	56.63	53.10	48.93	40.20	28.26
	NLPrompt	**62.97**	**61.23**	**59.17**	**55.17**	**49.03**	**39.80**	**62.30**	**60.60**	**56.47**	**50.80**	**40.27**	**28.37**
EuroSAT	CoOp	76.50	69.23	61.67	52.33	37.63	26.70	76.00	66.27	53.83	41.17	28.00	17.43
	GCE	82.13	78.60	74.67	63.13	49.67	31.40	78.23	72.70	63.63	45.30	22.90	12.10
	JoAPR	75.13	61.10	60.90	63.63	38.97	27.33	69.37	67.30	59.40	47.60	33.93	17.50
	NLPrompt	**82.53**	**79.53**	**78.13**	**66.70**	**63.53**	**43.80**	**80.13**	**77.13**	**71.43**	**54.30**	**37.27**	**32.73**
OxfordPets	CoOp	76.50	66.73	60.33	47.03	35.77	24.60	76.10	66.20	52.53	38.73	26.63	14.90
	GCE	85.63	84.60	83.67	79.23	71.40	53.17	85.50	83.03	76.73	68.07	50.70	31.97
	JoAPR	84.00	83.26	83.20	83.10	82.40	**74.40**	82.90	83.40	79.07	75.84	52.74	43.57
	NLPrompt	**86.17**	**86.00**	**85.33**	**84.87**	**83.63**	70.77	**86.00**	**84.97**	**82.40**	**77.53**	**66.33**	**48.60**
StanfordCars	CoOp	66.20	59.70	53.40	45.90	35.67	22.90	65.77	57.13	46.23	33.73	22.37	12.80
	GCE	**69.70**	66.40	66.47	63.77	59.25	50.87	**70.00**	66.45	61.23	53.67	39.65	26.60
	JoAPR	68.60	66.30	62.83	56.67	48.50	39.40	66.47	61.70	51.50	42.03	30.80	22.97
	NLPrompt	69.37	**68.80**	**67.20**	**65.63**	**62.83**	**58.30**	69.77	**67.53**	**64.23**	**59.03**	**50.90**	**39.50**
UCF101	CoOp	69.03	63.40	58.23	49.73	40.83	26.30	67.23	58.07	46.47	34.43	23.67	13.17
	GCE	74.00	**73.63**	72.57	69.37	66.00	57.07	73.90	71.87	67.97	62.23	52.50	36.37
	JoAPR	72.83	71.17	70.37	67.63	65.30	57.67	72.07	69.80	64.10	59.17	56.07	47.46
	NLPrompt	**74.83**	73.40	**72.83**	**70.33**	**68.10**	**60.53**	**74.90**	**73.53**	**71.03**	**65.97**	**58.97**	**49.27**
Caltech101	CoOp	86.43	81.03	76.73	70.90	61.33	46.90	84.93	75.23	62.87	49.43	33.57	20.33
	GCE	**92.00**	90.90	**90.80**	89.30	86.70	79.03	91.27	**91.20**	89.73	85.80	78.20	62.07
	JoAPR	90.30	90.45	89.90	88.27	86.93	83.93	90.30	89.30	88.30	88.73	85.80	81.90
	NLPrompt	91.73	**91.13**	90.77	**89.93**	**88.30**	**86.70**	**91.60**	91.17	**90.20**	**89.27**	**86.17**	**81.07**

Table 2. Test accuracy (%) on Food101N.

Method	CoOp	GCE	JoAPR	NLPrompt
Accuracy	69.50	71.32	72.57	76.46

13.27%. These improvements indicate that the OT split is not only filtering uniformly hard examples; it is also useful when corruption follows structured class transitions.

4.4. Results on Real Noisy Labels

Food101N contains naturally noisy web labels and is therefore a useful test of whether the method transfers beyond synthetic corruption. Table 2 shows that NLPromptreaches 76.46% accuracy, outperforming CoOp by 6.96 points, GCE by 5.14 points, and JoAPR by 3.89 points. The result indicates that the combination of prompt-level robust loss and OT purification is not tied to a hand-crafted corruption process.

5. Analysis

5.1. Why MAE Helps in Prompt Learning

Figure 2 directly compares CE and MAE during prompt learning on Caltech101. The trend is clear: MAE degrades much more gracefully as label corruption increases. This behavior matches Proposition 1. Once the prompt learner starts assigning low probability to suspicious labels, MAE sharply reduces the resulting gradients, preventing those examples from steering the prompt away from the pretrained CLIP geometry. CE does not have this safeguard, so noisy examples continue to exert large updates even when they are already inconsistent with the learned representation.

This observation also helps explain why prompt learning remains salvageable under extreme corruption. Because the image and text encoders are frozen, the prompt parameters only need to realign a relatively small contextual subspace. MAE prevents that subspace from being dominated by label outliers, while the OT split keeps a discriminative CE signal alive on the subset that still agrees with the image-text structure. The full method therefore behaves differently from applying a robust loss to a fully trainable classifier.

5.2. Ablation of PromptMAE and PromptOT

Table 3. Ablation studies under multiple label noise ratios on Flowers102. (%)

	Method/Noise Ratio	10%	30%	50%	70%	Avg
w\o OT	(a) all data with CE	92.71	86.92	79.36	57.61	79.15
	(b) all data with MAE	88.47	89.07	85.20	80.87	85.90
w\ OT	(c) w\o text feature init	87.16	83.81	79.77	73.00	80.94
	(d) w\o noisy data	84.77	84.53	81.60	77.60	82.16
	(e) w\o clean data	90.17	90.13	88.60	80.55	87.36
	NLPrompt	**96.87**	**93.44**	**92.30**	**85.38**	**92.00**

Table 3 studies the main components on Flowers102 under multiple noise ratios. Three conclusions follow. First, PromptMAE is a strong standalone choice: training all data with MAE improves the average accuracy from 79.15% to 85.90% over training all data with CE. Second, OT-based purification is necessary for the best results. Removing text-feature initialization inside the OT pipeline drops the average to 80.94%, showing that prompt-derived text prototypes are central to reliable partitioning. Third, both branches of the harmonized objective matter. Removing the noisy-data MAE branch reduces the average to 82.16%, while removing the clean-data CE branch gives 87.36%. The full method reaches 92.00%, which means the clean CE signal and the noisy MAE signal are complementary rather than interchangeable.

5.3. Effect of the Number of Shots

Figure 3 evaluates prompt tuning under different shot numbers. Although the exact values are visualized rather than tabulated in the provided artifacts, the trend is consistent: NLPromptimproves robustness throughout the few-shot regime instead of only at 16 shots. This matters because noisy-label sensitivity is often most damaging when only one or two mislabeled examples can distort an entire class prompt.

5.4. Failure Modes and Limits

The experimental record also reveals what NLPromptdoes *not* guarantee. It does not strictly dominate all alternatives in the low-noise regime, and it is not always the top method on every dataset under the hardest corruption. The OxfordPets and Caltech101 exceptions in Table 1 show that when the baseline prompt learner is already stable, or when a competing regularizer aligns particularly well with the dataset, the extra OT split may offer less benefit. This is a useful boundary condition: the value of NLPromptgrows with label corruption, but the method should not be sold as a universal drop-in improvement for nearly clean data.

The same results suggest a more specific mechanism behind these limits. PromptOT relies on the current prompt-induced text geometry, so its trusted subset is conservative when several classes remain semantically close or when early training has not yet separated them well. In those cases, the method still behaves safely because doubtful samples are routed to MAE instead of being reinforced by CE, but the clean subset can become too small to deliver the full discriminative gain of cross-entropy. This interpretation is consistent with the weaker margins on visually subtle datasets and in lower-noise settings, where avoiding wrong trust assignments matters more than aggressively expanding the clean pool.

Table 4. The generalization of NLPrompt .

Method/Noise Ratio	12.5%	25.0%	37.5%	50.0%	62.5%	75.0%
VPT	89.20	79.43	65.20	61.37	41.67	27.67
VPT+Ours	**91.80**	**91.07**	**89.53**	**86.93**	**80.73**	**73.90**
MaPLe	83.27	77.67	65.27	55.40	37.53	25.47
MaPLe+Ours	**89.23**	**84.30**	**78.37**	**76.43**	**73.30**	**59.87**
PromptSRC	90.61	84.67	78.57	72.27	60.43	49.37
PromptSRC+Ours	**91.29**	**87.67**	**84.97**	**80.33**	**72.10**	**59.50**

5.5. Transfer to Other Prompt Learners

Table 4 evaluates whether the proposed training recipe can be grafted onto other prompt learners. The answer is yes. Adding our clean/noisy harmonization to VPT improves accuracy from 27.67% to 73.90% at 75.0% noise, MaPLe from 25.47% to 59.87%, and PromptSRC from 49.37% to 59.50%. The gains persist across all listed noise ratios. This transfer result is important because it suggests that NLPromptis not tied to a particular prompt parameterization; instead, it acts as a general robust optimization layer for prompt-based adaptation.

6. Conclusion

We presented NLPrompt, a robust prompt-learning framework for noisy-label adaptation of vision-language models. The method combines two simple ideas: use MAE to attenuate low-consistency samples, and use prompt-induced text prototypes with entropic OT to identify which samples should still benefit from cross-entropy. The resulting clean/noisy harmonization consistently improves robustness across synthetic and real noisy-label benchmarks, with especially large gains under severe corruption. Taken together, the results position NLPromptas a practical robustification layer for prompt tuning: it extracts a trusted core when one exists, degrades gracefully when that core is small, and preserves the efficiency advantages of frozen-backbone adaptation. The ablations and transfer experiments further show that both PromptMAE and PromptOT are necessary and broadly compatible with other prompt-learning methods.

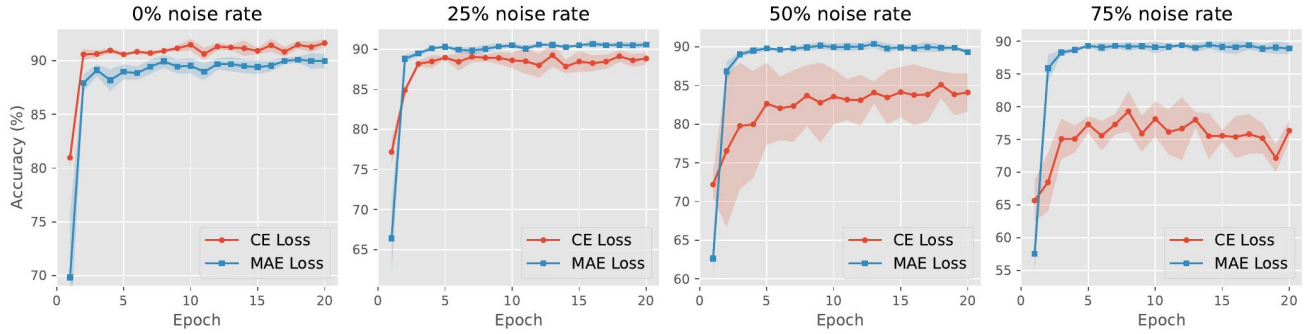

Figure 2. Training dynamics of CE and MAE prompt learning on Caltech101. The provided figure shows that MAE is markedly more stable than CE under noisy supervision, supporting the gradient attenuation analysis in Section 3.

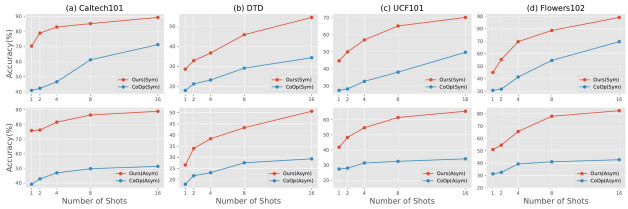

Figure 3. Accuracy under different numbers of shots. The provided figure shows that NLPromptremains effective from very low-shot settings to 16-shot training, indicating that the method is not dependent on a single supervision budget.

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